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Assignment - 1

Example

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Determine the absolute, relative and relative percentage errors when approximating P by P^* when $P = 0.37856 \times 10^1$ and $P^* = 0.379 \times 10^1$

Answer:

Absolute Error: $| p - p^* | \rightarrow | (0.37856 \times 10^1) - (0.379 \times 10^1) | = 4.4 \times 10^{-3}$

Relative Error: $| P - P^* | / | P | \rightarrow | 4.4 \times 10^{-3} | / | 0.37856 \times 10^1 | = 1.1623 \times 10^{-3}$

Relative percentage: $RE \times 100\% \rightarrow 1.1623 \times 10^{-3} \times 100\% = 1.1623 \times 10^{-3}$

Example

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Determine the absolute, relative and relative percentage errors when approximating P by P^* in each of the following cases:

Case	P	P^*
a	2.1666	2.1667
b	0.0004	0.0003
c	10000.0001	10000

Case	AE	RE	RPE
a	$\lvert 2.1666 - 2.1667 \rvert = 1 \times 10^{-4}$	$1 \times 10^{-4} / \lvert 2.1666 \rvert = 4.615 \times 10^{-5}$	$(4.615 \times 10^{-5}) \times 100\% = 4.1655 \times 10^{-5}$
b	$\lvert 0.0004 - 0.0003 \rvert = 1 \times 10^{-4}$	$1 \times 10^{-4} / \lvert 0.0004 \rvert = 0.25$	$(0.25) \times 100\% = 0.25$
c	$\lvert 10000.0001 - 1000 \rvert = 9 \times 10^{-3}$	$9 \times 10^{-3} / \lvert 10000.0001 \rvert\ 8.9 \times 10^{-7}$	$(8.9 \times 10^{-7}) \times 100\% = 8.9 \times 10^{-7}$